

The phase-dressed normal moment

Claude, with Daniel Murfet

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Abstract

Unit 201 (programme A, step 3). The one-dimensional normal moment dressed by a constant phase,

$$S_{p/2}^{(\beta)}(a) = \int_0^\infty s^{p-1} e^{-\beta s^2 + \beta a s} ds$$

(`phaseMoment` `p a`; the paper's $S_{p/2}(a)$, kernel `quadKernel` `a`), is formalised with everything the assembly unit needs: integrability with the uniform Gaussian domination $s^{p-1} e^{-\beta s^2 + \beta a s} \leq e^{\beta R^2/2} s^{p-1} e^{-\beta s^2/2}$ for $|a| \leq R$, continuity in a (dominated convergence on a ball), the Gaussian moments $\int_0^\infty s^{p-1} s^j e^{-\beta s^2} ds = \frac{1}{2} \Gamma(\frac{p}{2} + \frac{j}{2}) \beta^{-(p/2+j/2)}$, the zero-phase value $\frac{1}{2} \Gamma(p/2) \beta^{-p/2}$, and the *finite Taylor approximation*

$$\left| S_{p/2}^{(\beta)}(a) - \sum_{j < 2K} \frac{(\beta a)^j}{j!} \cdot \frac{1}{2} \Gamma\left(\frac{p}{2} + \frac{j}{2}\right) \beta^{-(p/2+j/2)} \right| \leq \tau_K(\beta, b) \int_0^\infty s^{p-1} e^{-\beta s^2/4} ds \quad (|\beta a| \leq b),$$

with the uniform tail constant τ_K of unit 199. Zero `sorry/axiom`; 9078 jobs.

1 The things formalised

```
noncomputable def phaseMoment ( p a : ) : :=
  s in Ioi 0, s ^ (p - 1) * quadKernel a s
noncomputable def gaussTail ( p : ) : := s in Ioi 0, s ^ (p - 1) * exp (-( /4 * s^2))
theorem gauss_integrableOn (hc : 0 < c) (hq : -1 < q) :
  IntegrableOn (fun s => s ^ q * exp (-(c * s^2))) (Ioi 0)
theorem quadKernel_le_of_abs_le (h) (hs : 0 ≤ s) (haR : |a| ≤ R) :
  quadKernel a s ≤ exp ( R^2/2) * exp (-( /2 s^2))
theorem phaseMoment_integrand_integrableOn (h) (hp : 0 < p) :
  IntegrableOn (fun s => s ^ (p - 1) * quadKernel a s) (Ioi 0)
theorem continuous_phaseMoment (h) (hp) : Continuous fun a => phaseMoment p a
theorem gaussMoment_eq (h) (hq : -1 < q) :
  s in Ioi 0, s ^ q * exp (-( s^2)) = Γ ((q+1)/2) * ^ (-((q+1)/2)) / 2
theorem gaussMoment_nat_eq (j : ) (h) (hp) :
  s in Ioi 0, s ^ (p-1) * (s ^ j * exp (-( s^2))) = Γ (p/2 + j/2) * ^ (-
(p/2 + j/2)) / 2
theorem phaseMoment_zero (h) (hp) : phaseMoment p 0 = Γ (p/2) * ^ (-p/2) / 2
theorem phaseMoment_taylor_le (h) (hp) (hab : |a| ≤ b) (K : ) :
  |phaseMoment p a - ∑ j in range (2K), (a)^j / j! * (Γ (p/2 + j/2) * ^(-
(p/2+j/2)) / 2)|
  tailCoeff b K * gaussTail p
```

2 Mathematics

The finite sum is rewritten as one integral, $\int_0^\infty s^{p-1} (\sum_{j < 2K} (s\beta a)^j / j!) e^{-\beta s^2} ds$, via the Gaussian moments (Mathlib's `integral_rpow_mul_exp_neg_mul_rpow` at $p = 2$); the difference with $S(a)$ is then a single integral whose integrand is bounded pointwise by `phase_tail_le`, and `norm_integral_le_of_norm_le` finishes. Nothing here depends on the exponent data (h, k) ; in the assembly the phase $a = \xi(\pi u)$ varies over the face and the bound is uniform because $|\beta\xi| \leq \beta\|\xi\|_\infty =: b$ on the cube.

3 Lean notes

- Dot notation `(h.const_mul c).congr_fun` fails with “`And.congr_fun`”: `Integrable` unfolds to a conjunction, so the result of `Integrable.const_mul` is not seen as an `IntegrableOn`. Ascribe the type in a `have` first.
- `integral_finset_sum/integrable_finset_sum` are deprecated for `integral_finsetSum/integrable_finsetSum`.
- `IntegrableOn.congr_fun` hands the pointwise goal as an applied lambda; `beta_reduce` before `rw`.
- `continuousAt_of_dominated` with the bound depending on the centre a_0 (radius-one ball, $R = |a_0| + 1$) is the clean way to get continuity of a parametric integral whose integrand is not globally dominated.

4 Next

Unit 202: the leading-term theorem. Define $C = \frac{1}{(m-1)! \prod_{i \in J} k_i} \int_{(0,1]^d} \eta(\pi u) S_{p/2}^{(\beta)}(\xi(\pi u)) w(u) du$ and prove $I_N / (N^{-p}(\log N)^{m-1}) \rightarrow C$ by `tendsto_of_approx`: the finite Taylor approximants converge by unit 200, the N -side error is bounded by the bare monomial theorem at $\beta/4$ times τ_K , and $|C - P_K|$ by `phaseMoment_taylor_le`.